CAMPUS CONNECT

"WHERE CAMPUS CONNECTS, ACADEMICS THRIVE"

Presented by Group 14
DevAxis

Github Repo Link

INTRODUCTION

A web-based platform bridging students and professors at IITK through structured feedback, project collaboration, and anonymous discussions

Lecture anonymous feedback

Project Management

Anonymous Discussions

Analytics Dashboard

NEED FOR THE PLATFORM

Informal & Fragmented Feedback
Lecture feedback is unstructured or collected through manual systems, making it hard for professors to analyze.

No Centralized Project Listings
Project opportunities are scattered across channel, No single platform for discovery and application.

Limited Anonymous Interaction
Students lack safe spaces for honest academic questions and campus-wide discussions.

No Data-Driven Teaching Insights
Professors cannot easily access visual analytics to evaluate teaching effectiveness or track trends over time.

POTENTIAL USERS

STUDENTS

Enroll in courses, ask doubts anonymously, apply for research projects, and submit session feedback , all from one dashboard

PROFESSORS

Create and manage courses, post research projects, review applicants, and get visual analytics on teaching performance

CAMPUS JUNTA

Post questions, share opinions, and discuss campus life, academics, placements, events, and more completely anonymously. No names, no judgment

MAJOR REQUIREMENTS

LECTURE FEEDBACK

- 5-parameter rating (Clarity, Pace, Engagement, Content, Delivery)
- Mandatory structured feedback per lecture
- Optional anonymous comments (500 char limit)
- Auto-save draft functionality
- Professor analytics: bar charts & trend lines

COURSE MANAGEMENT

- Professor creates courses with dates & schedule
- Auto-generate class sessions from date range
- Bulk student enrollment via Excel upload
- Manual add / remove students by email
- Co-instructor support with shared access

MAJOR REQUIREMENTS

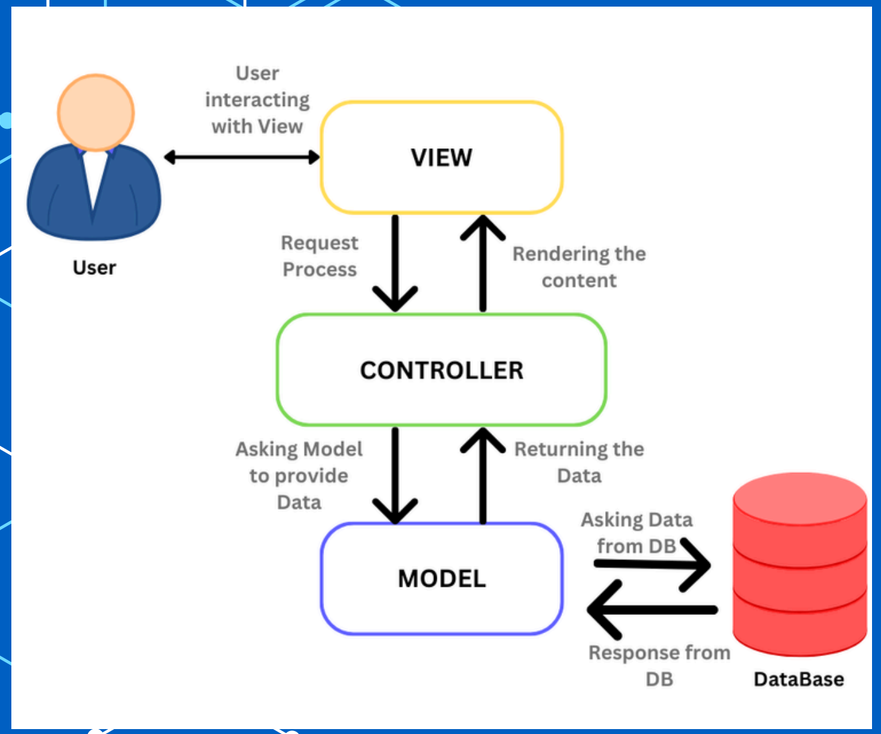
PROJECT MODULE

- Professors post projects with eligibility criteria (CPI, Branch)
- Students browse, filter, and apply with resume
- Application tracking: Pending / Accepted / Rejected
- Professor accepts/rejects with one click
- Project status: Open / Closed / Deleted

CAMPUS DISCUSSIONS

- Posts in categories: General, Academics, Projects, Study Group, Career
- All student posts are fully anonymous
- Professors post with actual name
- Threaded replies with upvote/downvote
- Tags, character limits, and draft auto-save

ARCHITECTURE DESIGN



MODEL

- User profiles (Students/Professors)
- Course details & sessions
- Project listings & applications
- Feedback forms & analytics data

VIEW

- Login / Signup pages
- Student & Professor dashboards
- Feedback forms & project listings
- Campus discussion forum

CONTROLLER

- Auth: Student & Professor login
- Route requests (Apply, Feedback, Post)
- Process feedback submissions
- Update model and refresh view

IMPLEMENTATION

FRONTEND : React 19 + Vite + Tailwind CSS
Responsive, component-driven UI with React Router DOM and Chart.js visualizations

BACKEND : Node.js + Express.js
RESTful API with JWT auth, Bcrypt hashing, Nodemailer OTP, and Socket.io real-time events

DATABASE : MongoDB + Mongoose
Document-based NoSQL storage with Mongoose ODM for flexible schema management

Real-Time : Socket.io 4.8
Live lecture discussion and real-time chat between students and professors

AUTH & EMAIL : JWT + Nodemailer
OTP-based registration and password recovery via IITK institutional email (SMTP)

DEMONSTARTION

WEBSITE LINK

FUTURE IMPLEMENTATIONS

- ML-BASED SENTIMENT ANALYSIS ON FEEDBACK COMMENTS FOR DEEPER INSIGHTS
- CLOUD DATABASE MIGRATION FOR SCALABILITY AND MULTI-INSTITUTE SUPPORT
- PDF & EMAIL GENERATION FOR COURSE REPORTS AND FEEDBACK SUMMARIES
- DEDICATED MOBILE APP FOR ANDROID AND iOS PLATFORMS
- REAL-TIME PUSH NOTIFICATIONS FOR DEADLINES AND ANNOUNCEMENTS

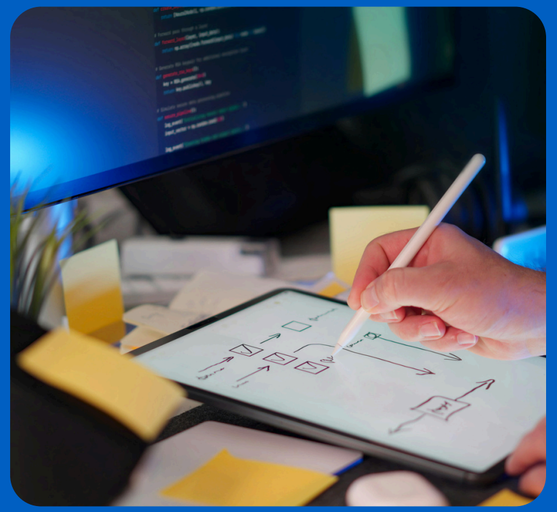
LEARNINGS...

OPERATIONAL

- Team Collaboration across 10 members
- Plan-driven Development Process
- Decision Making under constraints
- Time & Resource Management
- Accountability & Responsibility

TECHNICAL

- Version Control with Git & GitHub
- Frontend — React, Vite, Tailwind CSS
- Backend — Node.js + Express.js
- Database Handling using MongoDB
- Authentication & OTP via SMTP
- Deployment on cloud server



THANK YOU